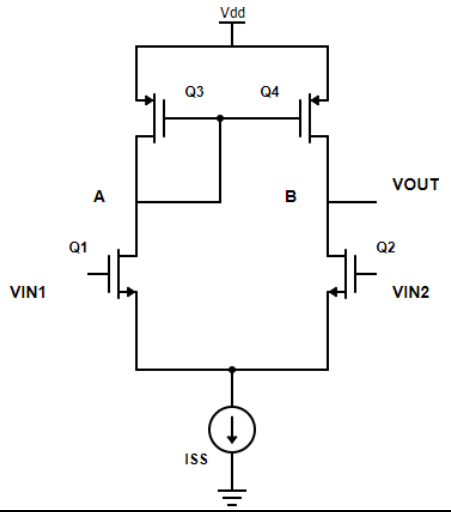
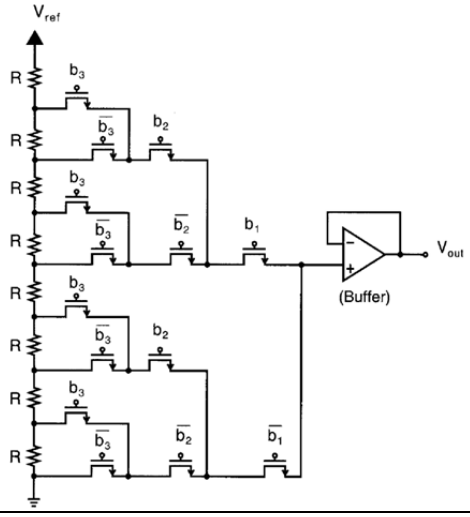
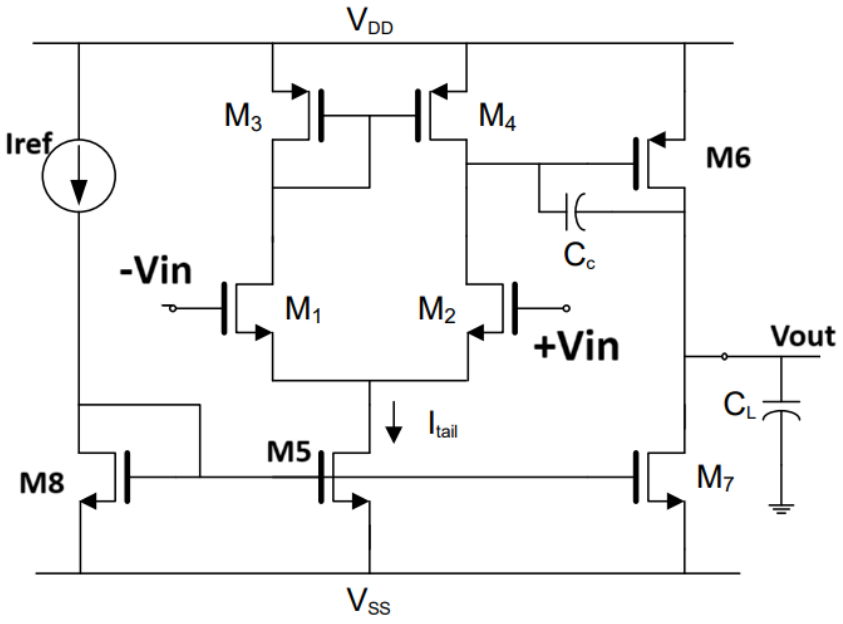
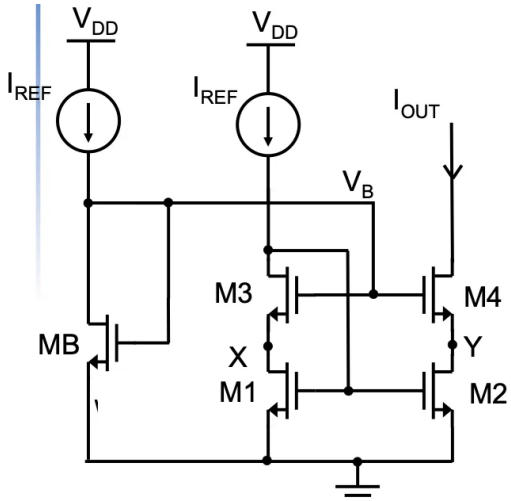
DEPT. OF ELECTRONICS AND COMMUNICATION ENGINEERING
END SEMESTER EXAMINATION NOV/DEC 2024 - QUESTION PAPER BLUEPRINT
ANALOG AND MIXED SIGNAL DESIGN– ECE-4061, SET-1

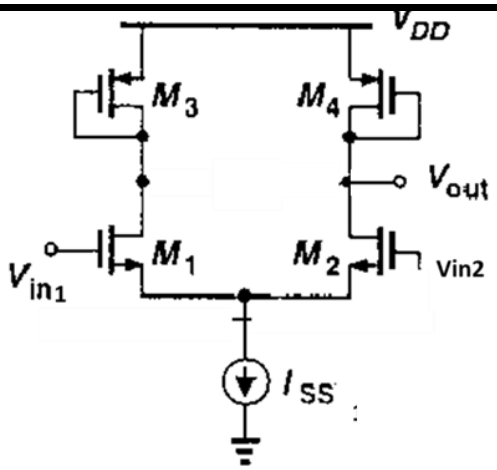
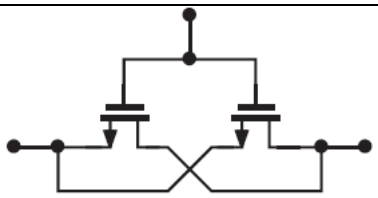
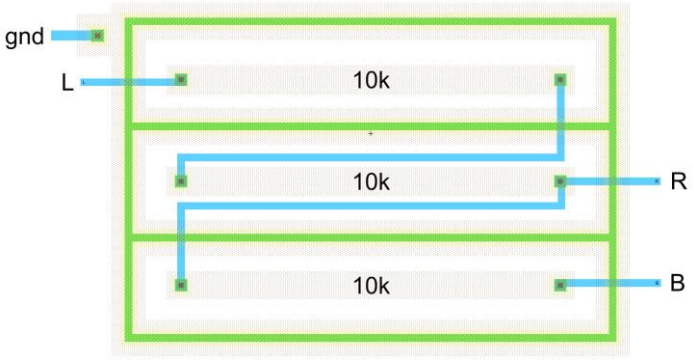
Question No	Topic	Marks
1a	Design a single stage OTA for a voltage gain of 20 and a power budget of 2 mW with Vdd=2 V. Assume Q1 operates at the edge of the saturation if common-mode level is 1V. Also $\mu n*Cox = 2\ \mu p*Cox = 100\ \mu A/V^2$, Vtn=0.5 V , Vtp= - 0.4 V, $\lambda n = 0.5\ V^{-1}$, $\lambda p = 0.1\ V^{-1}$. 	4
1b	A 3-bit binary DAC is shown with Vref=1V and R=100 Ω. Due to fabrication mismatch, 10% error has been introduced to all resistors. Calculate the maximum DNL for the DAC. 	3

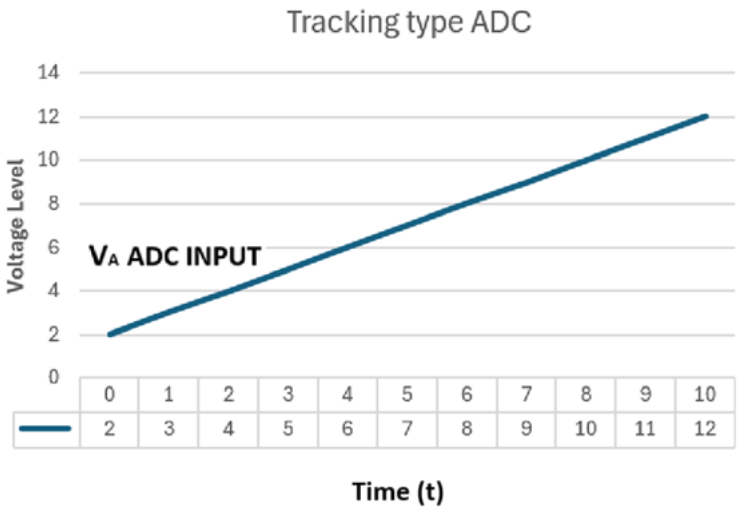
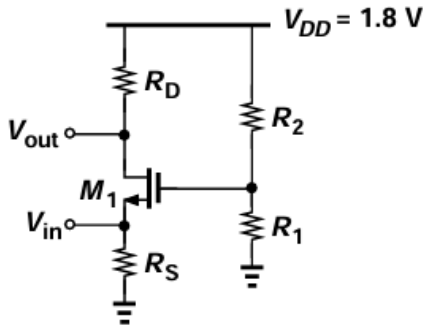
<table>
<tr>
<td>1c</td>
<td>Consider a set of simulated and ideal values obtained from DAC experiments as enumerated in Table below. Estimate the
(i) DNL and INL values.
(ii) DNL max and INL max
(iii) Plot the DNL and INL curve.
<table>
<tr><th>Simulated</th><th>Ideal</th></tr>
<tr><td>0</td><td>0</td></tr>
<tr><td>0.1907</td><td>0.1875</td></tr>
<tr><td>0.38131</td><td>0.375</td></tr>
<tr><td>0.57199</td><td>0.5625</td></tr>
<tr><td>0.7625</td><td>0.75</td></tr>
<tr><td>0.95317</td><td>0.9375</td></tr>
<tr><td>1.14381</td><td>1.125</td></tr>
<tr><td>1.33448</td><td>1.3125</td></tr>
<tr><td>1.52474</td><td>1.5</td></tr>
<tr><td>1.71542</td><td>1.6875</td></tr>
<tr><td>1.90605</td><td>1.875</td></tr>
<tr><td>2.09673</td><td>2.0625</td></tr>
<tr><td>2.28724</td><td>2.25</td></tr>
<tr><td>2.47792</td><td>2.4375</td></tr>
<tr><td>2.66855</td><td>2.625</td></tr>
<tr><td>2.85923</td><td>2.8125</td></tr>
</table>
</td>
<td>3</td>
</tr>
<tr>
<td>2a</td>
<td>Design a two stage OTA that meets the following specifications. Assume the channel length for all the transistors to be 1 µm.
Av>5000, Vdd=2.5V, Vss= -2.5 V, GB= 5MHz, CL=10 pF, SR> 10 V/µS, Voutrange= ±2V, PM= 60°, ICMR= -1 to 2V , Power dissipation ≤ 2 mW.

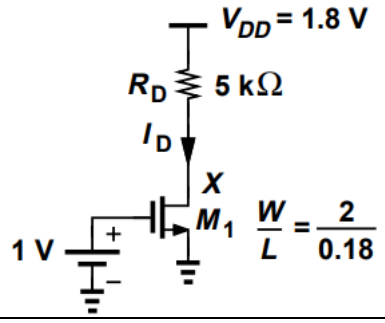
Nmos : Vt= 0.7 ± 0.15V, λn=0.04 V^{-1}, µn*Cox=110 µA/V^2 ± 10%

Pmos : Vt= -0.7 ± 0.15V, λp=0.05 V^{-1}, µn*Cox=50 µA/V^2 ± 10%</td>
<td>4</td>
</tr>
</table>

		
2b	Design a Wide Swing Cascode Current Mirror in 180-nm CMOS Technology. Hence, derive all required expressions. Given, Id= 100 µA, µn*Cox= 270 µA/V^2, Vt=0.45 V, Cox= 8.5 fF/µm^2, (λ*L)= 0.08 µm/V, L=1.6 µm, tox=5 nm. Estimate Rout. 	3
2c	Design an unbalanced differential amplifier with diode connected load with CLM and ac voltage source input with Gain= 250. Hence, derive all required expressions. Given, Iss= 200 µA, W/L=176, λ (pmos)= 0.05 V^{-1}, λ (nmos) = 0.1 V^{-1} , Vtn = 0.4V, Vtp =-0.4V , Vdd= 1.8V .	3

		
3a	 (i) Draw the layout of (ii) A n-well resistors are connected in serpentine configuration. Each unit of resistor is 10K. It has sheet resistance of 800 Ω and width of 12 µm. Estimate the Length (L) of the one resistor unit. 	4
3b	Explain the working of Tracking ADC with neat diagram. Trace the DAC output for given analog input.	3

		
3c	(i) An 8-bit ADC is to digitize a five-volt, full-scale signal. What is the accuracy with which the ADC converter can digitize the following signals, 1 V and 2.5 V? (ii) What is the largest value of output voltage from an 8-bit DAC that produces 1 V for a digital input of 00110010?	3
4a	Design the CG stage of the circuit such that it can accommodate an output swing of 500 mV p-p, i.e., Vout can fall below its bias value by 250 mV without driving M1 into the triode region. Assuming a voltage gain of 4 and an input impedance of 50 ohm and selecting R_S=10/gm and R1 +R2 =20kohm. 	4
4b	Calculate the bias current of M1 in the figure. Assume µn*Cox = 100 µA/V^2 and V_T = 0.4 V. If the gate voltage increases by 10 mV, what is the change in the drain voltage?	3

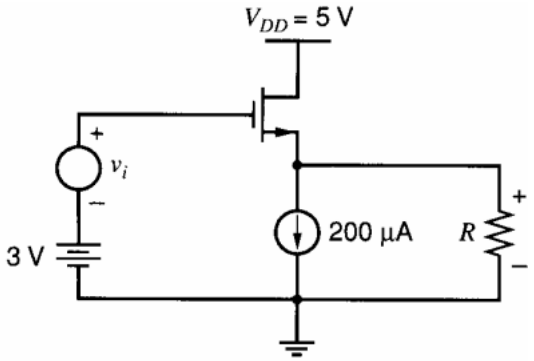


4c (3)

For common drain amplifier circuits, W/L=10, V_{t0}=0.7 V and λ=0. Estimate the dc output voltage Vo and small-signal gain for following conditions:

(i) Ignoring body effect with R=∞.

(ii) Including body effect with R=∞. Use Vsb= Vo (Vo from part (i)) , εr=11.7, gmb/gm= 0.2, V_T (thermal voltage) =26 mV, ni= 1.5x10^{10} /cm^3.



Summary of Process Parameters for a Typical Silicon-Gate *n*-Well CMOS Process with 1.5 µm Minimum Allowed Gate Length

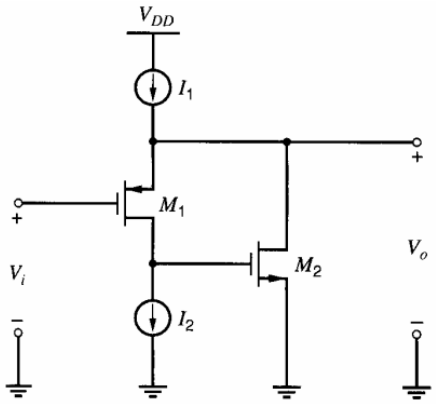
Parameter	Symbol	Value *n*-Channel Transistor	Value *p*-Channel Transistor	Units
Substrate doping	N_A, N_D	2×10^{15}	1.5×10^{16}	Atoms/cm^3
Gate oxide thickness	t_{ox}	250	250	Å
Metal-silicon work function	ϕ_{ms}	− 0.6	− 0.1	V
Channel mobility	μ_n, μ_p	650	300	cm^2/V-s
Minimum drawn channel length	L_{drwn}	1.5	1.5	µm

5a (4)

Estimate the C_{GB}, C_{GS} & C_{GD} for an MOS device that has a W of 5 µm and L of 1 µm for saturation region. Given,

Capacitance Values and Coefficients for the MOS Model

Type	n-Channel	Units
CGSO	220×10^{-12}	F/m
CGDO	220×10^{-12}	F/m
CGBO	700×10^{-12}	F/m
CJ	770×10^{-6}	F/m^2
CJSW	380×10^{-12}	F/m
MJ	0.5	
MJSW	0.38	Based on an oxide thickness of 140 Å or $C_{ox} = 24.7 \times 10^{-4}$ F/m^2.

5b	(i) A MOSFET carries a drain current of 1 mA with V_{DS} = 0.5 V in saturation. Determine the change in ID if V_{DS} rises to 1 V and λ= 0.1 V^{-1}. What is the device output impedance? (ii) Assuming λ α 1/L, calculate ΔI_D and ro in question 5b(i), if both W and L are double.	3
5c	For the circuit shown below, derive the expressions for Rout and Av with body effect and CLM. M1=PMOS, M2=NMOS. 	3